

PRCL-0019 — Quick Summary

Sales Effectiveness — Lead Category Prediction for FicZon Inc.

The Problem

FicZon's sales leads were sorted manually into good/bad categories, and quality checks only happened after the fact. The goal: use ML to flag each lead as High Potential or Low Potential the moment it's captured, so agents know where to focus first.

The Data

7,422 real leads pulled live from FicZon's MySQL database (9 columns: source, agent, location, delivery mode, status, etc.). Missing values were kept as an "Unknown" category rather than dropped; 2 duplicate rows were removed.

The Key Challenge

The target column (Lead_Category) had to be built from the raw Status field. An early mapping used guessed status labels that didn't exactly match the database's real, case-sensitive values (e.g. 'CONVERTED' vs. 'Converted'), silently collapsing every lead into one class. The fix: inspect the real status strings and rebuild the mapping from them — restoring a genuine 61.6% / 38.4% Low/High split.

The Models

Five algorithms were compared, each deliberately regularized to avoid overfitting on the sparse, 64-column, one-hot-encoded feature set:

Model	Test Acc.	F1-Score	Overfit Gap
Gradient Boosting (best)	72.04%	0.7112	2.70 pts
KNN	69.00%	0.6810	5.24 pts
Logistic Regression	68.13%	0.6851	2.71 pts
Random Forest (Tuned)	67.72%	0.6816	6.69 pts

Gradient Boosting won on every metric that matters, including having the smallest train-test gap — meaning it generalizes best rather than just memorizing training data.

What Drives Predictions

Geography (location) is the single strongest signal, followed by the product and delivery mode. Sales-agent identity and customer-referral leads matter too.

Business Takeaways

- Score leads automatically at capture so agents prioritize the right ones immediately.
- Conversion likelihood varies strongly by region — consider territory-based strategy.
- Rebalance marketing spend toward higher-converting sources.
- Standardize status entry (fixed dropdown) to prevent future case-sensitivity bugs.

Bottom Line

A tuned Gradient Boosting model reaches 72% accuracy with the best generalization of six models tested — a solid, deployable starting point for prioritizing leads, not a finished, fully automated system.